

Phishing URL Detection Lab Report

This lab, I was focused on analysing characteristics commonly associated with suspicious URLs and developed a python-based phishing URL scorer. The detector assigns a risk score based on specific URL characteristics.

A python script was created using `urllib.parse` module and `re` module. The script analyses each URL and assigns points when suspicious characteristics are detected.

The 10 URLs samples that were tested, including legitimate looking domains and URLs containing characteristics commonly associated with phishing attempts.

URLs that do not begin with `https` receive 30 points because `https` encrypts communication between the browser and website but alone, it does not provide that the website is legitimate.

The detector checks the domain for keywords such as `login`, `verify`, `secure`, `update`, `account`, `bank` and `paypal`. These words can appear in legitimate websites, so this factor is only an indicator and does not automatically mean that the URL is malicious.

If a domain contains more than 3 dots, the detector adds 25 points. Attackers can sometimes use multiple subdomains to make a suspicious domain appear more convincing. EXAMPLE: “`paypal.loginevil.example.com`”, so the important part to identify is the actual registered domain rather than simply trusting the beginning of the URL.

If the hostname contains an IPv4, the detector adds 40 points. Using an IP address instead of a recognisable domain can be a suspicious characteristic. Example: <http://192.168.1.100/login>, this receives 70 points because it also used the `http` (30 points)

In closure

The phishing URL scorer successfully identified suspicious characteristics in the sample URLs and assigned risk scores according to the defined rules.